GRACE ONESIME ZONGO

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SUMMARY

Electrical Engineering student (B.S. expected May 2027) with three internships across embedded systems, HVAC controls, and power electronics. Co-author of a WAMC 2026 presentation on NSF-supported Antarctic weather-station firmware.

EDUCATION

University of Wisconsin, Madison **September 2024 – present**

- BS: Electrical Engineering: Expected May 2027
- UW Credit Union Scholar and JCKF scholar

Madison College, Madison, WI **January 2022 – May 2024**

- Associate degree: Liberal Arts Transfer: Science / Math / Technology
- Dean's List – Perfect Honors for all 5 semesters.

TECHNICAL SKILLS

Programming: C/C++, Python, VHDL/Verilog, MATLAB

Tools/Hardware: Altium, KiCad, LTspice, Intel Quartus (FPGA), Oscilloscope/DMM, DDC Controllers, Git/GitHub.

Concepts: Embedded Systems, Power Electronics, Low-Power Design, Finite State Machines, USART/I²C/SPI/CAN

INTERNSHIP EXPERIENCES

AWS Student Intern | Space Science and Engineering Center – UW Madison | Madison, WI **April 2024 – Present**

- Reduced system power consumption by 78% and improved runtime efficiency by 85% by implementing a custom low-power sleep mode and C++ Finite State Machine.
- Co-authored a WAMC 2026 presentation (NSF-supported); the station ran unattended in Antarctica for a full year.
- Created adaptive frequency sampling method for sensor drivers, and tested sensors for calibration.

Design Engineer Intern | Alpha Controls and Services | Middleton, WI **May 2025 – December 2025**

- Analyzed HVAC mechanical blueprints to perform material takeoffs, select sensors, actuators, and Schneider controllers.
- Created electrical schematics and wiring diagrams for HVAC systems using equipment submittals and sequence of operation.
- Prepared cost estimates for projects by quantifying control system components and materials.

Electrical Engineering Intern | Bemis Manufacturing Company | Madison, WI **June 2022 – August 2022**

- Designed a prototype of Bemis's first smart, health monitoring bidet seat for household use.
- Integrated an optical heart-rate sensor into the seat, with a mechanism to record its health data.

PROJECT EXPERIENCE

Digitally Controlled Buck Converter (personal project) **2026**

- Built a synchronous DC-DC buck converter with a UART-programmable 1.8–10 V output; bare-metal C on STM32G431KB with 200 kHz PWM, ADC + DMA sensing, and a 10 kHz PID loop with anti-windup and overcurrent protection.
- Laid out the PCB in Altium, verified the power stage in LTspice, and debugged bring-up faults on the bench.

Promoting Electric Propulsion (PEP) competition for electric boats **September 2023 – April 2024**

- Designed a dual 18650 Li-ion battery system and differential thrust control for an electric catamaran, contributing to a 2nd place finish at the PEP 2024 Competition.

Honors Project: Designing an 8-bit CPU on Quartus **June 2023 – December 2023**

- Designed and simulated an 8-bit register, RAM, ALU from scratch to implement Von Neumann architecture.
- Implemented FETCH, LOAD, and ADD instructions, and designed binary to BCD converter for segment displays.

LEADERSHIP EXPERIENCE

Wisconsin Idea Fellow | Tech'Up Jeunes Initiative | UW–Madison **May 2026 – May 2027**

- Design and lead a hands-on electronics program for high-school students in Burkina Faso with Leading Change Africa.

Clubs Development Coordinator | Executive Leadership Team (ELT) | Madison College **March 2022 – May 2024**

- Supported 50+ student clubs, administered the campus-wide club platform, and organized officer training.